Mathematical Methods for Physics—Lecture 19

Classification and Separation of Variables

Instructor: Zhaolong Han, SOCE, Shanghai Jiao Tong University
han.arkey@sjtu.edu.cn

After classifying the basic equations, we turn from whole-line initial-value problems to finite-interval mixed problems. We first determine the allowed spatial modes of a string and then use the same eigenfunctions for wave, heat, and potential equations. The interval and the boundary conditions matter at every stage.

1 Classification of Second-Order Equations

Consider a linear equation in two independent variables whose principal part is

$$Au_{xx} + 2Bu_{xy} + Cu_{yy} + \text{lower-order terms} = f. \tag{1}$$

At a point where A, B, C do not all vanish, the characteristic-direction equation is

$$A(\,\mathrm{d}y)^2 - 2B\,\mathrm{d}x\,\mathrm{d}y + C(\,\mathrm{d}x)^2 = 0. \tag{2}$$

Its discriminant is $4(B^2 - AC)$. Equivalently, use $D = B^2 - AC$:

Type	Criterion	Basic example
Hyperbolic	$D > 0$	$u_{tt} - a^2u_{xx} = 0$
Elliptic	$D < 0$	$u_{xx} + u_{yy} = 0$
Parabolic	$D = 0$	$u_t - \kappa u_{xx} = 0$

The wave and heat examples use (x, t) rather than (x, y) as independent variables. If the coefficient of u_{xy} is denoted by B instead of $2B$, the corresponding discriminant is $B^2 - 4AC$.

2 A String with Two Fixed Ends

2.1 Separation and the spatial eigenvalue problem

Let $a > 0$, $\ell > 0$, and consider the mixed problem

$$\begin{cases} u_{tt} - a^2u_{xx} = 0, & 0 < x < \ell,\ t > 0, \\ u(0,t) = u(\ell,t) = 0, & \\ u(x,0) = \varphi(x), \qquad u_t(x,0) = \psi(x). & \end{cases} \tag{3}$$

Seek a nonzero separated solution $u(x,t) = X(x)T(t)$. Then

$$XT'' - a^2X''T = 0, \qquad \frac{X''}{X} = \frac{T''}{a^2T} = -\lambda. \tag{4}$$

The ratios are used where division is possible; the resulting ODEs hold also at zeros by continuity:

$$X'' + \lambda X = 0, \qquad T'' + a^2\lambda T = 0, \qquad X(0) = X(\ell) = 0. \tag{5}$$

The spatial problem asks for the values of λ that admit a nonzero X. Such values are *eigenvalues*, and the corresponding functions are *eigenfunctions*.

Case $\lambda = 0$. Here $X = C_1 + C_2 x$. Both endpoint values vanish only for $C_1 = C_2 = 0$, so zero is not an eigenvalue for this boundary pair.

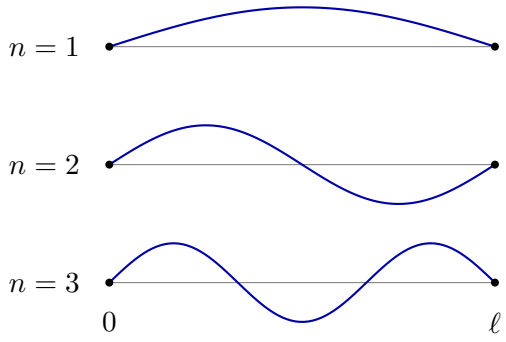
Case $\lambda = -\mu^2 < 0$. Write $X = C_1 \cosh(\mu x) + C_2 \sinh(\mu x)$. The left condition gives $C_1 = 0$; the right then gives $C_2 = 0$ because $\sinh(\mu \ell) > 0$.

Case $\lambda = k^2 > 0$. Now $X = C_1 \cos(kx) + C_2 \sin(kx)$. We obtain $C_1 = 0$ and $\sin(k\ell) = 0$. Therefore

$$\boxed{k_n = \frac{n\pi}{\ell}, \quad \lambda_n = k_n^2, \quad X_n(x) = \sin(k_n x), \quad n = 1, 2, \dots.} \tag{6}$$

Set $\omega_n = a k_n$. The corresponding time factor is

$$T_n(t) = A_n \cos(\omega_n t) + B_n \sin(\omega_n t). \tag{7}$$



Each shape satisfies the endpoint conditions, but one shape alone generally cannot reproduce the prescribed initial displacement.

2.2 Orthogonal basis and Fourier sine expansion

For real functions on $(0, \ell)$, define the inner product

$$\langle v, w \rangle = \int_0^\ell v(x) w(x) \, \mathrm{d}x. \tag{8}$$

The sine eigenfunctions are orthogonal:

$$\int_0^\ell \sin(k_n x) \sin(k_m x) \, \mathrm{d}x = \begin{cases} 0, & n \neq m, \\ \ell/2, & n = m. \end{cases} \tag{9}$$

They form a *complete orthogonal basis* in $L^2(0, \ell)$. Thus a square-integrable function can be approximated by finite sums of these modes, with mean-square error tending to zero. This is the Fourier sine expansion. Completeness permits the expansion; orthogonality determines its coefficients.

For example, writing

$$\varphi_N(x) = \sum_{n=1}^{N} A_n \sin(k_n x) \tag{10}$$

is analogous to resolving a vector into orthogonal directions. Multiplication by $\sin(k_m x)$ and integration isolate A_m. Infinite expansions require a convergence statement; they do not follow from linearity alone.

2.3 Superposition and determination of coefficients

Finite sums of the separated solutions solve the PDE and homogeneous boundary conditions. Using the complete spatial family gives the series representation

$$u(x,t) = \sum_{n=1}^{\infty} [A_n \cos(\omega_n t) + B_n \sin(\omega_n t)] \sin(k_n x). \tag{11}$$

At $t = 0$, the initial conditions require

$$\varphi(x) = \sum_{n=1}^{\infty} A_n \sin(k_n x), \qquad \psi(x) = \sum_{n=1}^{\infty} \omega_n B_n \sin(k_n x). \tag{12}$$

Orthogonality gives

$$\boxed{A_n = \frac{2}{\ell} \int_0^{\ell} \varphi(x) \sin(k_n x)\, \mathrm{d}x, \qquad B_n = \frac{2}{\ell \omega_n} \int_0^{\ell} \psi(x) \sin(k_n x)\, \mathrm{d}x.} \tag{13}$$

The factor $1/\omega_n$ in B_n comes from differentiating $\sin(\omega_n t)$ once in time.

The procedure is a method of undetermined coefficients in a spatial orthogonal basis: first find the eigenfunctions, then solve the time equations, and finally use the initial data to determine their amplitudes. For energy solutions the initial expansions are interpreted in the corresponding integral norms. Pointwise satisfaction of the PDE requires enough regularity to differentiate the series and satisfy the endpoint compatibility conditions.

3 Examples for Fixed Ends

3.1 Example 10.1: a polynomial initial displacement

Consider the initial data of Example 10.1 (textbook p.262):

$$\varphi(x) = \frac{x(\ell - x)}{M}, \qquad \psi(x) = 0, \qquad M > 0. \tag{14}$$

If an amplitude constant C is used instead, it must be $C = 1/M$. We immediately have $B_n = 0$. Two integrations by parts give

$$\int_0^{\ell} x(\ell - x) \sin(k_n x)\, \mathrm{d}x = \frac{1}{k_n} \int_0^{\ell} (\ell - 2x) \cos(k_n x)\, \mathrm{d}x = \frac{2[1 - (-1)^n]}{k_n^3}. \tag{15}$$

Hence

$$A_n = \frac{4\ell^2}{M n^3 \pi^3} [1 - (-1)^n]. \tag{16}$$

Even coefficients vanish, and the Fourier-series solution is

$$u(x,t) = \frac{8\ell^2}{M\pi^3} \sum_{j=0}^{\infty} \frac{\cos((2j+1)\pi a t/\ell) \sin((2j+1)\pi x/\ell)}{(2j+1)^3}. \tag{17}$$

The odd-mode selection also reflects the symmetry of the initial shape about $x = \ell/2$. *Regularity remark.* This example defines an energy solution; additional corner compatibility would be needed for a globally C^2 classical solution.

3.2 Initial data containing only two modes

Take

$$\varphi(x) = \sin\frac{\pi x}{\ell}, \qquad \psi(x) = \sin\frac{4\pi x}{\ell}. \tag{18}$$

Orthogonality gives $A_1 = 1$, $B_4 = \ell/(4\pi a)$, with all other coefficients zero. Therefore

$$u(x,t) = \cos\frac{\pi a t}{\ell}\sin\frac{\pi x}{\ell} + \frac{\ell}{4\pi a}\sin\frac{4\pi a t}{\ell}\sin\frac{4\pi x}{\ell}. \tag{19}$$

When the initial data contain finitely many eigenfunctions, the general infinite series reduces to finitely many terms.

4 How Endpoint Conditions Select the Spectrum

4.1 Four standard endpoint pairs

For $X'' + \lambda X = 0$ on $(0,\ell)$, the positive eigenvalues are $\lambda_n = k_n^2$:

Endpoint conditions	k_n, $n \geq 1$	$X_n(x)$
$X(0) = X(\ell) = 0$	$n\pi/\ell$	$\sin(k_n x)$
$X(0) = X'(\ell) = 0$	$(2n-1)\pi/(2\ell)$	$\sin(k_n x)$
$X'(0) = X(\ell) = 0$	$(2n-1)\pi/(2\ell)$	$\cos(k_n x)$
$X'(0) = X'(\ell) = 0$	$n\pi/\ell$	$\cos(k_n x)$

For the Neumann–Neumann pair there is also $\lambda_0 = 0$, $X_0 = 1$. The squared norm is $\ell/2$ for each displayed mode and ℓ for X_0. As an example, $X(0) = 0$ removes the cosine, while $X'(\ell) = 0$ requires $\cos(k\ell) = 0$, which explains the half-integer wave numbers.

4.2 Example 10.2: one fixed and one free end

For $u(0,t) = 0$, $u_x(\ell,t) = 0$, let

$$\varphi(x) = 3\sin\frac{3\pi x}{2\ell} + 6\sin\frac{5\pi x}{2\ell}, \qquad \psi(x) = 0. \tag{20}$$

These are the second and third modes of the Dirichlet–Neumann family. Consequently,

$$u(x,t) = 3\cos\frac{3\pi a t}{2\ell}\sin\frac{3\pi x}{2\ell} + 6\cos\frac{5\pi a t}{2\ell}\sin\frac{5\pi x}{2\ell}. \tag{21}$$

4.3 Two free ends and the zero mode

For $u_x(0,t) = u_x(\ell,t) = 0$, the expansion is

$$u(x,t) = A_0 + B_0 t + \sum_{n=1}^{\infty}[A_n\cos(\omega_n t) + B_n\sin(\omega_n t)]\cos(k_n x), \tag{22}$$

where $k_n = n\pi/\ell$ and $\omega_n = a k_n$. For $\lambda_0 = 0$, the time equation is $T_0'' = 0$, so $T_0 = A_0 + B_0 t$. This term must not be discarded. The coefficients are

$$A_0 = \frac{1}{\ell}\int_0^\ell \varphi(x)\,\mathrm{d}x, \qquad B_0 = \frac{1}{\ell}\int_0^\ell \psi(x)\,\mathrm{d}x, \tag{23}$$

$$A_n = \frac{2}{\ell}\int_0^\ell \varphi(x)\cos(k_n x)\,\mathrm{d}x, \qquad B_n = \frac{2}{\ell\omega_n}\int_0^\ell \psi(x)\cos(k_n x)\,\mathrm{d}x \quad (n \geq 1). \tag{24}$$

The mean displacement is $A_0 + B_0 t$, describing uniform transverse translation.

5 The Same Spatial Modes for Heat Conduction

For $u_t - \kappa u_{xx} = 0$, separation with $X'' + \lambda X = 0$ gives

$$T' + \kappa\lambda T = 0, \qquad T_n(t) = C_n e^{-\kappa\lambda_n t}. \tag{25}$$

The alternative heat coefficient a^2 is denoted here by κ, as in Lecture 18. With insulated endpoints $u_x(0,t) = u_x(\ell,t) = 0$,

$$u(x,t) = C_0 + \sum_{n=1}^{\infty} C_n e^{-\kappa(n\pi/\ell)^2 t} \cos\frac{n\pi x}{\ell}, \tag{26}$$

where

$$C_0 = \frac{1}{\ell}\int_0^\ell \varphi(x)\,\mathrm{d}x, \qquad C_n = \frac{2}{\ell}\int_0^\ell \varphi(x)\cos\frac{n\pi x}{\ell}\,\mathrm{d}x. \tag{27}$$

The zero mode is constant because $T_0' = 0$. All positive modes decay. In agreement with this,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_0^\ell u(x,t)\,\mathrm{d}x = \kappa[u_x(\ell,t) - u_x(0,t)] = 0. \tag{28}$$

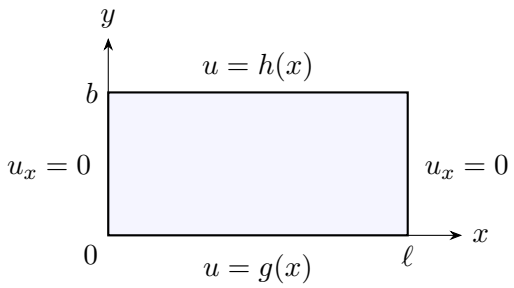
The long-time temperature approaches the initial spatial average. For endpoints held at zero temperature, use the sine family and omit the zero mode.

6 Laplace's Equation in a Rectangle

6.1 Spatial separation and the zero mode

Consider

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x < \ell,\ 0 < y < b, \\ u_x(0,y) = u_x(\ell,y) = 0, & \\ u(x,0) = g(x), \qquad u(x,b) = h(x). & \end{cases} \tag{29}$$



Put $u = X(x)Y(y)$. Then

$$X'' + \lambda X = 0, \quad X'(0) = X'(\ell) = 0, \qquad Y'' - \lambda Y = 0. \tag{30}$$

The spatial modes are $X_0 = 1$ and $X_n = \cos(k_n x)$, $k_n = n\pi/\ell$. For the remaining coordinate,

$$Y_0 = \alpha_0 + \beta_0 y, \qquad Y_n = \alpha_n \cosh(k_n y) + \beta_n \sinh(k_n y) \quad (n \geq 1), \tag{31}$$

where $\cosh z = (e^z + e^{-z})/2$ and $\sinh z = (e^z - e^{-z})/2$. The variable in the zero mode is y, and $Y_0'' = 0$ must be solved separately.

6.2 Matching the two remaining sides

Define cosine coefficients for the boundary data:

$$g_0 = \frac{1}{\ell}\int_0^\ell g(x)\,\mathrm{d}x, \qquad h_0 = \frac{1}{\ell}\int_0^\ell h(x)\,\mathrm{d}x, \tag{32}$$

$$g_n = \frac{2}{\ell}\int_0^\ell g(x)\cos(k_n x)\,\mathrm{d}x, \qquad h_n = \frac{2}{\ell}\int_0^\ell h(x)\cos(k_n x)\,\mathrm{d}x \quad (n \geq 1). \tag{33}$$

Matching at $y = 0$ and $y = b$ gives

$$\alpha_0 = g_0, \quad \beta_0 = \frac{h_0 - g_0}{b}, \qquad \alpha_n = g_n, \quad \beta_n = \frac{h_n - g_n\cosh(k_n b)}{\sinh(k_n b)}. \tag{34}$$

Thus an equivalent form of the complete solution is

$$u(x,y) = g_0\left(1 - \frac{y}{b}\right) + h_0\frac{y}{b} + \sum_{n=1}^{\infty}\left[g_n\frac{\sinh(k_n(b-y))}{\sinh(k_n b)} + h_n\frac{\sinh(k_n y)}{\sinh(k_n b)}\right]\cos(k_n x). \tag{35}$$

The positive modes are harmonic and have zero x derivative on the vertical sides. The two hyperbolic-sine factors attain the lower and upper boundary values respectively. The mean mode interpolates linearly between g_0 and h_0. Smooth solutions up to corners require compatible boundary data; for less regular data the boundary expansions are interpreted through their appropriate traces.

One Spatial Problem, Three Different Evolutions

For the same spatial equation $X'' + \lambda X = 0$ and endpoint pair,

PDE	Remaining equation	Zero-eigenvalue factor
Wave	$T'' + a^2\lambda T = 0$	$A_0 + B_0 t$
Heat	$T' + \kappa\lambda T = 0$	C_0
Laplace	$Y'' - \lambda Y = 0$	$\alpha_0 + \beta_0 y$

The spatial boundary conditions select the modes; the PDE determines their time or remaining-coordinate factors; the data determine the coefficients. The next lecture extends this procedure to nonzero boundary data and forcing.

Textbook pointers. Separation of variables: p.258 onward. Example 10.1: p.262. Endpoint spectra and Example 10.2: Table 10-1, p.265. Exercises: p.292, Problem 1(1)–(4) and Problem 2.